Imperial College London

Chip Design projects

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URL: https://elec70142-digital-vlsi-design.github.io/course-website/
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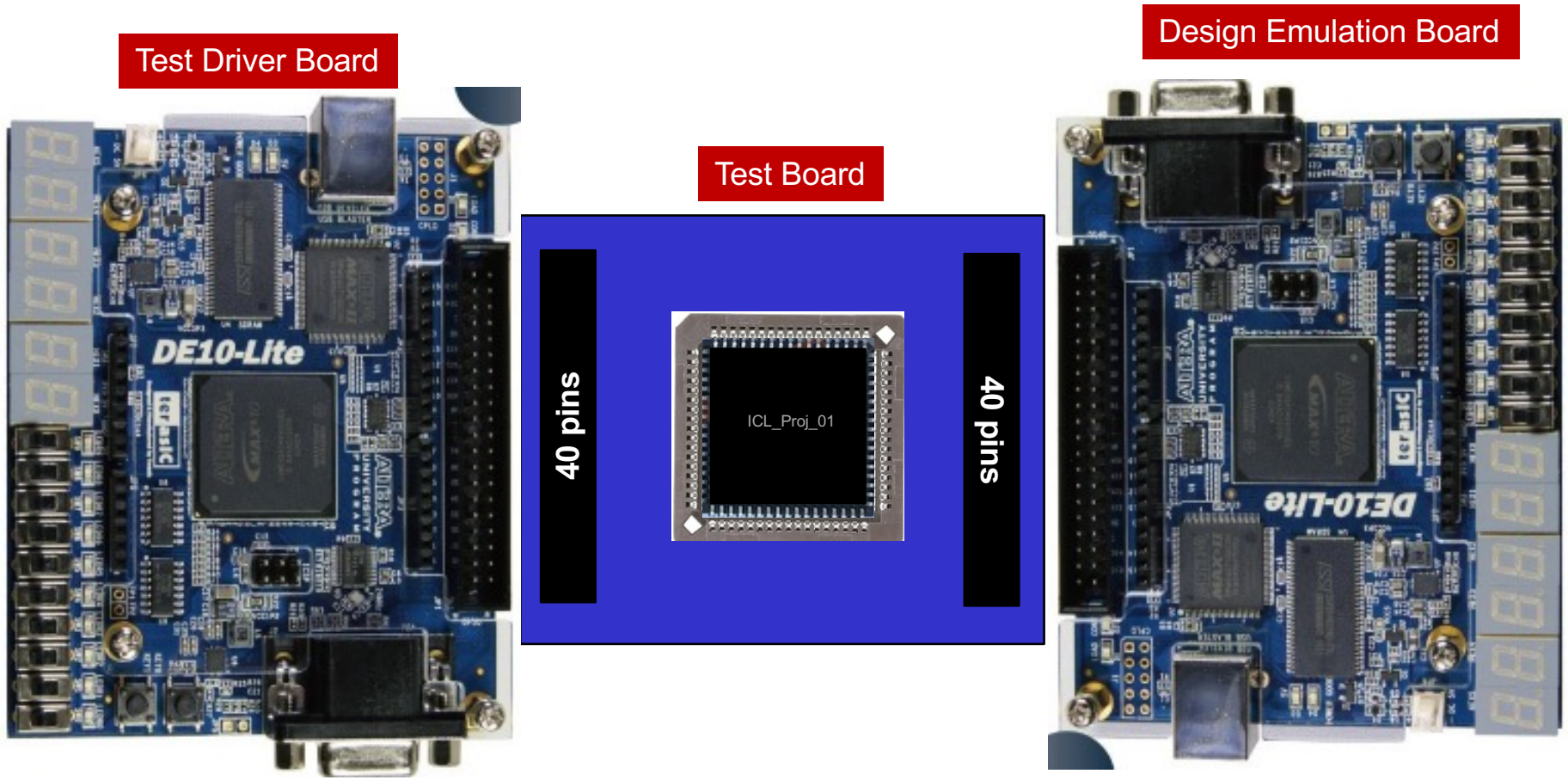
What will we cover today

- Key dates
- Test Environment
- Design Constraints
- Integration Guidelines
- Phase Lock Loop and Clocking issues

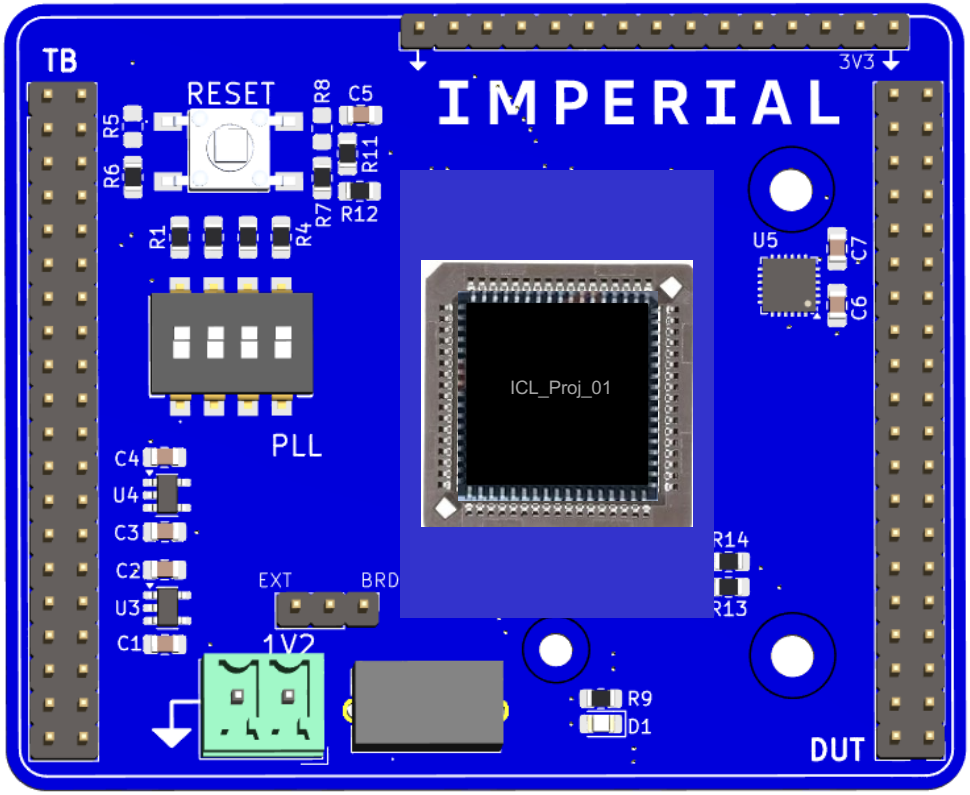
Important Dates

- **22 Oct** – **Official start** of team chip design projects
- **9 November** – Create and submit Project Github repos by team's repo master
  - Add pykc, cowardsa, shl2019, Mayank-Manas and bardia01 to repo for read-only access
  - Complete the Team's repo URL form via this link:
    ELEC70142 Team Repo Links – Fill in form
- **11 November** – Team submits Project Specification via Github
- **12 November** – Progress review with academic and industrial advisors
- **2 December** – Peter releases **Project Template and Deliverable requirements**
- **10 January** 23.59 – Deadline for Team Project via GitHub
- **17 February** – TSMC Tapeout
- **Early June 2027** – Fabricated chip arrive
- **Mid-June 2027** – Chip testing

Test Environment

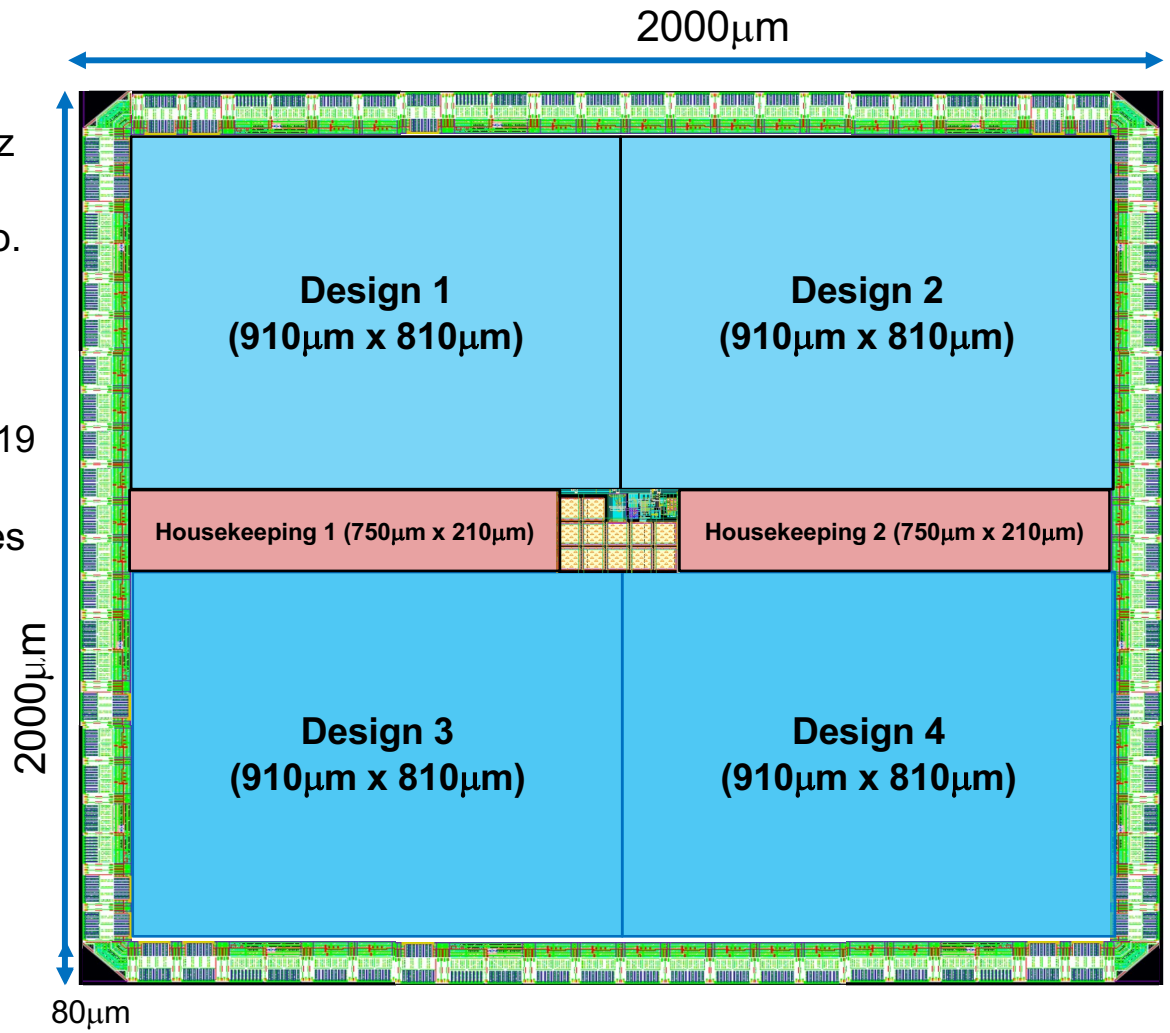


Test Board


TB
RESET
IMPERIAL
3V3
R5
R6
R8
C5
R11
R7
R12
R1
R4
U5
C7
C6
ICL_Proj_01
PLL
C4
U4
C3
C2
U3
C1
EXT
BRD
1V2
R14
R13
R9
D1
DUT

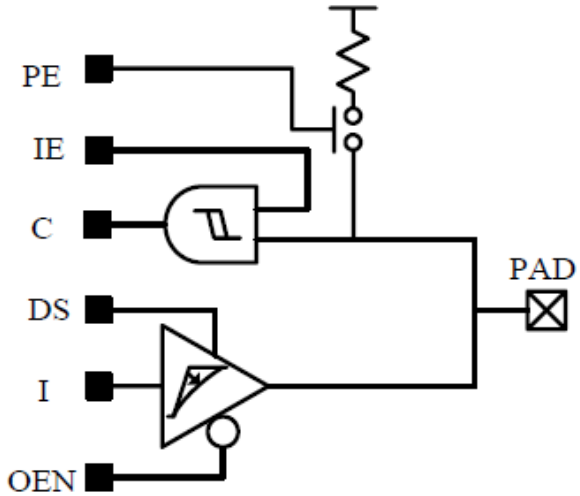
Design Constraints

Area:	910 μm x 810 μm
Clock Frequency:	From 25 MHz to 400 MHz
Clock pin:	CLKx – x is the design no.
Input pins:	16 pins INx_0 to INx_15
Output pins:	20 pins OUTx_0 to OUTx_19
Technology:	TSMC 65 – digital libraries used during the labs



Input / Output

- **Digital IO cell: PRUW0408SCDG**
- **Each digital IO cell provides:**
  - **Input and output enable controls**
  - **Schmitt Trigger Input**
  - **Output drive strength for**
  - **Pull-up resistor**
- **More information and truth tables in the documentation**
- **Since 4 designs share I/O pin, IO cells will be constrained to be INPUT or OUTPUT only.**



Pin Name	Direction	Description
PE	Input	Pull-up enable
IE	Input	Input enable
C	Input	Input signal
DS	Input	Driving Strength
I	Output	Output signal
OEN	Input	Output Enable

Resources
IO Cells Datasheet
/usr/local/synopsys/kits/tsmc/beLibs/65nm/TSMCHOME/digital/Documentation/documents/tphn65lpnv2od3_sl_200b/DB_TPHN65LPNV2OD3_SL_TC.pdf

IO Application Note
/usr/local/synopsys/kits/tsmc/65n_LP/doc/LIBS/TSMC Slim IO General Application Note_1_.pdf

Verification & Testing

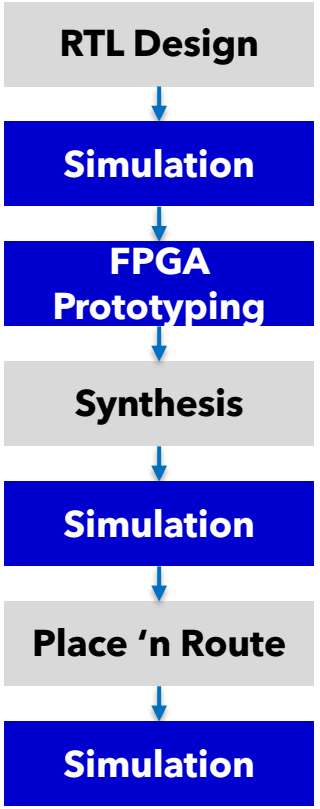
Before Synthesis

- **Simulation**
  - **Verilator (C++)**
  - **Cocotb (Python)**
  - **Vivado/Quartus (SystemVerilog)**
  - **Synopsys VSC (Simulator)**
- **Testing**
  - **Prototype on DE10 Lite**

Post Synthesis/Layout

- **Synopsys VSC (Simulator)**

Verification takes time!



Communication Protocols & Peripherals

Communication Interfaces for loading data, debugging, or connecting peripherals.

- **UART – Data exchange or debugging with a PC or .**
- **SPI – for sensors or memory devices.**
- **I2C –for low-speed configuration or sensors.**

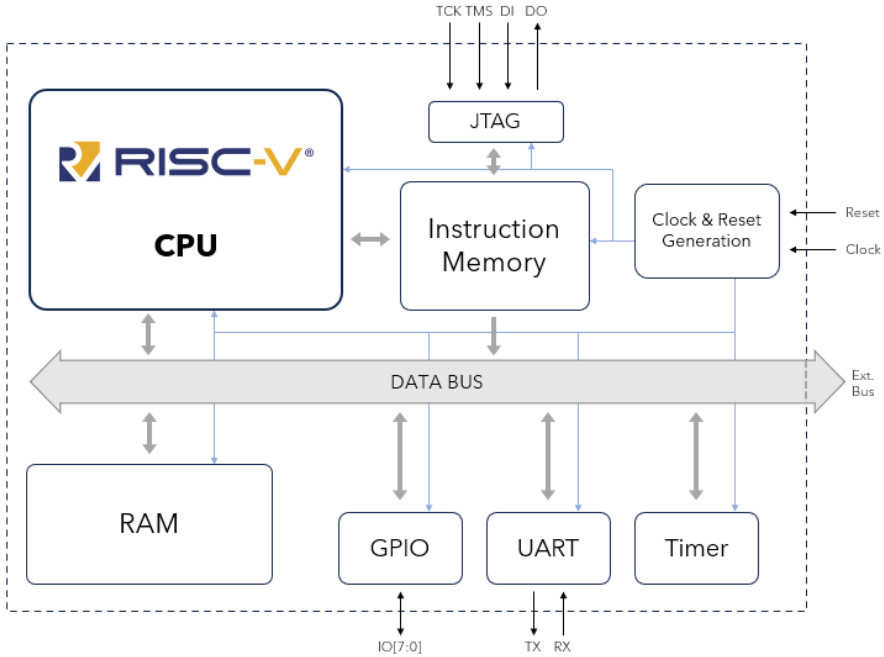
Available Peripherals on DE10-Lite Board

- **Switches & Buttons.**
- **LEDs & 7-Segment Displays.**
- **Accelerometer (I^2C interface)**
- **SDRAM (64 MB) – *requires custom interface*.**
- **VGA Output – *consider number of pins required*.**

Compute Options

You can add a CPU to your project.
The web offers several Open-Source CPU Cores and SoC:

- **Yosis PicoRV32**
- **ETH PULP Platform**
- **CV32E40P**



… or a simpler SoC written in SystemVerilog implementing a RV32IM CPU with memory-mapped GPIO, UART and Timer peripherals.

- **https://github.com/ridoluc/rvcpu-rv32im-soc**

Memory Options

On-Chip

- **SRAM (from compiler)**
  - ***Preferred option* for implementing larger memories.**
  - **Typically used for instruction and data memory, buffers, caches, lookup tables.**
- **Flip-Flops (Synthesized Memory)**
  - ***Use for small memories or register files.***
  - ***Synthesized from standard cells — larger area.***
- **Hardcoded (as ROM)**
  - ***For small lookup tables, coefficients, or initialization values.***
  - ***Implemented from HDL, synthesized using Tie-High/Low cells.***

Off-Chip (DE10-Lite Board)

- **SDRAM**
  - ***64MB SDRAM External memory available.***
  - ***Requires custom interface due to limited IOs.***
- ***FPGA Flip-Flops***
  - ***For simulating external memory***

On-Board Memory Initialization

During Simulation

- **Use Verilog initial blocks or `$readmemh` / `$readmemb` to preload memory.**
- **Convenient for instruction or weight initialization during functional verification.**

In the Design

- **Option 1 – JTAG Memory Interface**
  - **Custom controller allowing *read/write access* via JTAG instructions.**
  - **Useful for loading instruction memory or model weights at runtime.**
  - **Reference implementation and testbenches:**
    github.com/ridoluc/jtag-memory-controller

- **Option 2 – Custom Data Loader**
  - **Implement a simple protocol for loading data through UART, SPI, or other I/O.**
  - **More effort, but flexible and project-dependent.**